

An Improved Bound for Colorings Without Symmetrically Colored Four-Term Progressions

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Abstract

Under a coloring c , a nontrivial four-term arithmetic progression (4-AP) $a, a + d, a + 2d, a + 3d$ is called *symmetrically colored* if $c(a) = c(a + 3d)$ and $c(a + d) = c(a + 2d)$. Deng, Tidor, and Zhao asked whether $[N]$ admits a coloring with $N^{o(1)}$ colors and no such 4-AP, and gave an $O(N^{\log_{22} 3})$ -coloring of $[N]$. We give an explicit $27p$ -coloring of $\mathbb{Z}/p^4\mathbb{Z}$ for every prime $p \geq 5$ and hence an $O(N^{1/4})$ -coloring of $[N]$. The construction combines a carry-control coloring of base- p digits with the map $(u, t) \mapsto N_{\mathbb{F}_{p^3}/\mathbb{F}_p}(u) + t$. Together with Behrend-style product colorings, this gives $h(N) \leq N^{1/4+o(1)}$ in Erdős's Problem 160 on coloring every nontrivial 4-AP with at least three colors. This result also yields $\rho_4(\alpha) = O_\varepsilon(\alpha^{5-\varepsilon})$ for every $\varepsilon > 0$, improving the bound toward Ruzsa's question.

1 Introduction

Given a coloring c , we say that a four-term arithmetic progression (4-AP)

$$a, \quad a + d, \quad a + 2d, \quad a + 3d$$

is *nontrivial* if $d \neq 0$, and *symmetrically colored* if

$$c(a) = c(a + 3d) \quad \text{and} \quad c(a + d) = c(a + 2d).$$

Deng, Tidor, and Zhao [DTZ23, Conjecture 1.6] proposed the following Ramsey-type problem: study colorings of $[N] = 1, 2, \dots, N$ without nontrivial symmetrically colored 4-APs, and conjectured that $N^{o(1)}$ colors suffice. Their tensor-power construction, based on a three-coloring of $\mathbb{Z}/22\mathbb{Z}$, yields

$$O\left(N^{\log_{22} 3}\right),$$

where $\log_{22} 3 \approx 0.355$. To the best of our knowledge, this is the strongest previously known upper bound for this problem. Our main result improves this to $O(N^{1/4})$ via an explicit construction.

Theorem 1.1. *For every prime $p \geq 5$, there is a $27p$ -coloring of $\mathbb{Z}/p^4\mathbb{Z}$ with no nontrivial symmetrically colored 4-APs.*

This has two immediate applications. The first concerns Erdős Problem 160 [Erd89]:

Let $h(N)$ be the smallest k such that $1, \dots, N$ can be colored with k colors so that every 4-AP contains at least three distinct colors. Estimate $h(N)$.

Hunter [Hun25] observed that the standard Behrend-style product coloring eliminates all relevant two-color patterns except the symmetric pattern. Combining this observation with our result improves the exponent in the previously known upper bound from $0.355 + o(1)$ to $1/4 + o(1)$.

Corollary 1.2 (An improved upper bound for Erdős Problem 160). *One has*

$$h(N) = O(N^{1/4+o(1)}) .$$

Second, let $\rho_4(\alpha)$ denote the minimum asymptotic density of 4-APs in a Fourier-uniform set of density α , in the sense of [DTZ23, Definition 1.3]. Ruzsa [CL07, Problem 3.2] asked whether $\rho_4(\alpha)$ decays faster than every power of α . Combining the reduction of [DTZ23] with our result improves the exponent in the previously known upper bound from 4.406 to arbitrarily close to 5.

Corollary 1.3 (An improved upper bound for Ruzsa's question). *For every $\varepsilon > 0$, one has*

$$\rho_4(\alpha) = O_\varepsilon(\alpha^{5-\varepsilon}) \quad (0 < \alpha < 1/2) .$$

2 From 4-APs to digitwise 4-APs

Fix a prime $p \geq 5$. For $x \in 0, 1, \dots, p-1$, define

$$\tau(x) := \left\lfloor \frac{3x}{p} \right\rfloor \in 0, 1, 2 .$$

Thus τ divides $0, 1, \dots, p-1$ into three consecutive intervals, each of diameter less than $p/3$.

We first show the fact that makes the construction insensitive to carries.

Lemma 2.1. *Let $x_0, x_1, x_2, x_3 \in \{0, 1, \dots, p-1\}$ satisfy*

$$x_{i+1} \equiv x_i + s \pmod{p} \quad (i = 0, 1, 2)$$

for some $s \in \{0, \dots, p-1\}$. For $i = 0, 1, 2$, define the carry

$$e_i := \mathbf{1}_{\{x_{i+1} < x_i\}} .$$

If $\tau(x_0) = \tau(x_3)$ and $\tau(x_1) = \tau(x_2)$, then $e_0 = e_1 = e_2$.

Proof. If $s < 2p/3$, then, since $\tau(x_1) = \tau(x_2)$, necessarily $s < p/3$. If $s \geq 2p/3$, we instead consider the reversed order $(x'_0, x'_1, x'_2, x'_3) = (x_3, x_2, x_1, x_0)$ with $s' = p - s < p/3$, and

$$e'_i := \mathbf{1}_{\{x'_{i+1} < x'_i\}} = 1 - \mathbf{1}_{\{x_{3-i} < x_{2-i}\}} \quad (i = 0, 1, 2) .$$

Thus it suffices to consider the case $s < p/3$.

Let $\tau(x_1) = \tau(x_2) = i$. Since the step is less than $p/3$, the point x_0 lies either in interval i or $i-1$, while x_3 lies either in interval i or $i+1$ (interval indices understood modulo 3). As $\tau(x_0) = \tau(x_3)$, they must both lie in interval i . Therefore $e_0 = e_1 = e_2 = 0$. $\square$

Lemma 2.2 (digitwise 4-AP). *Let $L \geq 2$, and let*

$$n_k = a + kd \in \mathbb{Z}/p^L\mathbb{Z} \quad (k = 0, 1, 2, 3) .$$

For each $n \in \mathbb{Z}/p^L\mathbb{Z}$, write its standard representative as

$$n = \sum_{j=0}^{L-1} x_j(n)p^j, \quad x_j(n) \in 0, 1, \dots, p-1,$$

and set

$$\mathbf{x}(n) := (x_0(n), \dots, x_{L-1}(n)) \in \mathbb{F}_p^L.$$

Suppose that for every $j = 0, \dots, L-2$,

$$\tau(x_j(n_0)) = \tau(x_j(n_3)) \quad \text{and} \quad \tau(x_j(n_1)) = \tau(x_j(n_2)).$$

Then there is an $\mathbf{h} \in \mathbb{F}_p^L$ such that

$$\mathbf{x}(n_k) = \mathbf{x}(n_0) + k\mathbf{h} \quad (k = 0, 1, 2, 3).$$

Moreover, if $d \neq 0$ in $\mathbb{Z}/p^L\mathbb{Z}$, then $\mathbf{h} \neq 0$.

Proof. Iterating Lemma 2.1 from digit 0 to $L-2$ shows that the three carries entering each digit are equal, and hence that the four digits in every position form a 4-AP in $\mathbb{F}_p$; collecting the common differences gives $\mathbf{h}$, while $\mathbf{h} = 0$ would imply $n_0 = n_1$ and hence $d = 0$. $\square$

3 The cubic norm construction

Choose an irreducible cubic polynomial $f(T) \in \mathbb{F}_p[T]$, and let α be one of its roots in $\mathbb{F}_{p^3}$. Then $\{1, \alpha, \alpha^2\}$ is an $\mathbb{F}_p$ -basis of $\mathbb{F}_{p^3}$. Identifying $(x_1, x_2, x_3) \in \mathbb{F}_p^3$ with $x_1 + x_2\alpha + x_3\alpha^2$, define

$$P(x_1, x_2, x_3) := N_{\mathbb{F}_{p^3}/\mathbb{F}_p}(x_1 + x_2\alpha + x_3\alpha^2),$$

where

$$N_{\mathbb{F}_{p^3}/\mathbb{F}_p}(z) := z^{1+p+p^2}.$$

The existence of $f(T)$, as well as the fact that P is a homogeneous cubic polynomial with no nontrivial zero, is standard in the theory of finite fields; see, for example, [LN96, Theorem 2.28].

We shall use the following standard finite-difference identity.

Lemma 3.1. *For all $\mathbf{u}, \mathbf{v} \in \mathbb{F}_p^3$,*

$$P(\mathbf{u} + 3\mathbf{v}) - 3P(\mathbf{u} + 2\mathbf{v}) + 3P(\mathbf{u} + \mathbf{v}) - P(\mathbf{u}) = 6P(\mathbf{v}).$$

Proof. The polynomial $g(t) := P(\mathbf{u} + t\mathbf{v})$ has degree at most three in t . Since P is homogeneous of degree three, the coefficient of t^3 in g is $P(\mathbf{v})$. The third forward difference of a cubic polynomial is six times its leading coefficient, which gives the identity. $\square$

4 The norm-plus-linear construction

We write a vector in $\mathbb{F}_p^4$ as $(\mathbf{u}, t)$, where $\mathbf{u} \in \mathbb{F}_p^3$ and $t \in \mathbb{F}_p$, and define

$$Q(\mathbf{u}, t) := P(\mathbf{u}) + t.$$

Lemma 4.1 (norm plus linear). *Let*

$$\mathbf{y}_k = \mathbf{y}_0 + k\mathbf{h} \quad (k = 0, 1, 2, 3)$$

be a 4-AP in $\mathbb{F}_p^4$. If $Q(\mathbf{y}_0) = Q(\mathbf{y}_3)$ and $Q(\mathbf{y}_1) = Q(\mathbf{y}_2)$, then $\mathbf{h} = 0$.

Proof. Write

$$\mathbf{y}_0 = (\mathbf{u}, t), \quad \mathbf{h} = (\mathbf{v}, s)$$

with $\mathbf{u}, \mathbf{v} \in \mathbb{F}_p^3$ and $t, s \in \mathbb{F}_p$. By combining the two symmetric equalities with Lemma 3.1, we have

$$\begin{aligned} 0 &= Q(\mathbf{y}_3) - 3Q(\mathbf{y}_2) + 3Q(\mathbf{y}_1) - Q(\mathbf{y}_0) \\ &= (P(\mathbf{u} + 3\mathbf{v}) + 3s) - 3(P(\mathbf{u} + 2\mathbf{v}) + 2s) + 3(P(\mathbf{u} + \mathbf{v}) + s) - P(\mathbf{u}) \\ &= P(\mathbf{u} + 3\mathbf{v}) - 3P(\mathbf{u} + 2\mathbf{v}) + 3P(\mathbf{u} + \mathbf{v}) - P(\mathbf{u}) \\ &= 6P(\mathbf{v}). \end{aligned}$$

Since $p \geq 5$, we obtain $P(\mathbf{v}) = 0$. Since P has no nontrivial zero, we have $\mathbf{v} = 0$.

Now $P(\mathbf{u} + \mathbf{v}) = P(\mathbf{u} + 2\mathbf{v}) = P(\mathbf{u})$. Hence $Q(\mathbf{y}_1) = Q(\mathbf{y}_2)$ reduces to

$$t + s = t + 2s,$$

so $s = 0$. Therefore $\mathbf{h} = 0$. □

Proof of Theorem 1.1. For $n \in \mathbb{Z}/p^4\mathbb{Z}$, write its standard base- p expansion as

$$n = x_0(n) + px_1(n) + p^2x_2(n) + p^3x_3(n).$$

Set

$$\mathbf{u}(n) := (x_0(n), x_1(n), x_2(n)) \in \mathbb{F}_p^3$$

and define

$$c_4(n) := (\tau(x_0(n)), \tau(x_1(n)), \tau(x_2(n)), Q(\mathbf{u}(n), x_3(n))).$$

This coloring uses at most $3^3p = 27p$ colors.

Suppose that $n_k = a + kd$, $k = 0, 1, 2, 3$, is symmetrically colored by c_4 . Applying Lemma 2.2 with $L = 4$, we have

$$\mathbf{x}(n_i) = \mathbf{x}(n_0) + i\mathbf{h}$$

for some $\mathbf{h} \in \mathbb{F}_p^4$. Lemma 4.1 therefore implies that $\mathbf{h} = 0$. Lemma 2.2 then gives $d = 0$ in $\mathbb{Z}/p^4\mathbb{Z}$. Hence there is no nontrivial symmetrically colored 4-AP. □

Corollary 4.2. *For every positive integer N , the interval $[N]$ admits a coloring with $O(N^{1/4})$ colors and no symmetrically colored 4-AP. More precisely, for all sufficiently large N , at most $54N^{1/4}$ colors suffice.*

Proof. By Bertrand's postulate, for every positive integer $N \geq 81$, there is a prime $p \geq 5$ with

$$\lfloor N^{1/4} \rfloor < p < 2\lfloor N^{1/4} \rfloor.$$

Restricting the coloring from Theorem 1.1 to $[N] \subseteq [p^4]$ gives the result. □

5 Application to Erdős Problem 160

In Erdős Problem 160, one must exclude not only the symmetric $ABBA$ pattern, but all color patterns on a nontrivial 4-AP that use at most two colors. Hunter [Hun25] observed that the remaining patterns can be eliminated by taking the product with a Behrend-style coloring [Beh46] using $O(N^{o(1)})$ colors. We offer a proof of this observation, based on arguments of [DTZ23, Lemmas 3.2 and 7.11].

Lemma 5.1 (Behrend-style coloring). *There is a coloring*

$$\psi_N : [N] \longrightarrow [N^{o(1)}]$$

such that, if a nontrivial 4-AP receives at most two colors, then its color sequence is genuinely two-colored ABBA.

Proof of Corollary 1.2. Let c_N be the coloring from Corollary 4.2 and let ψ_N be the coloring in Lemma 5.1. Color $n \in [N]$ by $(c_N(n), \psi_N(n))$. This uses $O(N^{1/4+o(1)})$ colors. $\square$

6 Consequence for Fourier-uniform sets

Deng, Tidor, and Zhao [DTZ23] proved that an r -coloring of $\mathbb{Z}/M\mathbb{Z}$ without symmetrically colored 4-APs implies

$$\rho_4(\alpha) = O_{r,M}(\alpha^{3+\frac{1}{2}\log_r M}) , \tag{1}$$

for $0 < \alpha < 1/2$; see [DTZ23, Theorem 1.8].

Proof of Corollary 1.3. By Theorem 1.1, the exponent in (1) may be taken to be

$$3 + \frac{1}{2} \log_{27p}(p^4) = 3 + \frac{2 \log p}{\log p + \log 27} .$$

This tends to 5 as $p \rightarrow \infty$. $\square$

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